Problem A. Boys don't cry!

Input file:	*standard input*
Output file:	*standard output*
Time limit:	2 seconds
Memory limit:	512 mebibytes

Laska is the son of the king of the toilet. The pressure of being the son of such an important person was too big for him, so he decided to ask himself one really important question – what do I really want to do with my life? He answered himself that he want to be high all the time.

While being high he got amazing vision of permutations. They were forming different shapes, different patterns. Out of those infinite mix of permutations, he was able to distinguish n permutations of integers $1, 2, \ldots, m$.

He picked one of those permutations, let it be $\pi_1, \pi_2, \ldots, \pi_m$. He created a new permutation based on this with following algorithm:

- Start with empty sequence
- Analyze elements in order $1, 2, \ldots, m$. For element π_i he appended it either to the beginning or to the end of current sequence.

For example if initial permutation was $(5, 2, 3, 1, 4)$, he could create permutation $() \to (5) \to (2, 5) \to (3, 2, 5) \to (3, 2, 5, 1) \to (4, 3, 2, 5, 1)$.

He applied this operation exactly once to each of his initial permutations. Surprisingly it turned out that all resulting permutations were the same.

The next day he told you this story. Now you wonder how many different permutations could be created in such way. Moreover you want to restore the lexicographically minimal possible permutation which could be created in that way (if such exists).

As a reminder permutation $a_1, a_2, \ldots, a_m$ is lexicographically smaller then permutation $b_1, b_2, \ldots, b_m$ if there exist an index k fulfilling $a_k < b_k$ and $a_i = b_i$ for $1 \le i < k$.

Input

In the first line one integer $Z \le 50$ is given, denoting number of testcases described in following lines.

The first line of the each test case contains two integers n and m, denoting the number of permutations and length of those permutations. Each of the following n lines contains a description of those permutations. A description of a permutation consists of m pairwise distinct integers from set $\{1, 2, \ldots, m\}$. $n, m \in [1, 1000]$, sum of n over all testcases does not exceed 5 000

Output

For each test case, in the first line your program should write the remainder of number of permutations that could be created by $10^9 + 7$. If there exist at least one such permutation then in the second line you should write lexicographically minimum such permutation in the same format as in the input. If there is no such permutation then you should output only one line.

Example

standard input	standard output
4 1 5 1 2 3 4 5 2 5 1 2 5 4 3 5 2 1 4 3 2 5 1 2 3 4 5 5 4 3 2 1 3 5 1 2 3 4 5 1 3 5 2 4 3 4 2 1 5	16 1 2 3 4 5 8 1 2 5 4 3 2 1 2 3 4 5 0

Problem B. The Witcher

Input file:	*standard input*
Output file:	*standard output*
Time limit:	6 seconds
Memory limit:	512 mebibytes

The Witcher is now in some serious trouble! He needs to prepare a pocket map for his upcoming journey. He is now in a tavern and here there is a map of the whole Continent. For simplicity, we assume that the map is just an undirected graph without loops, but it may contain multiple edges between the same pair of nodes. There is an unwritten rule in the Witcher's world, that each graph placed in a pocket map should satisfy the following condition: the degree of each vertex should be even. The Witcher doesn't want to break the rule and also the graph in the pocket map should have the same set of nodes as the graph from the map in the tavern and the edges of this graph should form a subset of the edges from the graph in the tavern. Of course, there are some edges, which are necessary for the Witcher to complete his upcoming adventure, so he would also like to place them in his pocket map. The Witcher would like to know, if he can construct a valid graph, fulfilling the given conditions and if he can, then you should tell him, which edges he should take into his pocket map.

Input

In the first line one integer $Z \leq 100$ is given, denoting number of testcases described in following lines.

The first line of the standard input contains two integers n, m meaning the number of nodes and the number of edges in the graph from the map in the tavern. Each of the next m lines contains the description of one edge in the graph. i-th of them contains three integers a_i, b_i, x_i meaning that the i-th edge is connecting the nodes with numbers a_i and b_i, and if $x_i = 1$, then the Witcher needs this edge in his pocket map, otherwise you can choose it, but it is not necessary.

($n \in [1, 5 \cdot 10^5]$, $m \in [0, 5 \cdot 10^5]$), ($a_i, b_i \in [1, n]$, $a_i \neq b_i$, $x_i \in \{0, 1\}$).

Output

For each test case, the first line of the standard output should contain one word "`TAK`", if the Witcher can prepare the pocket map fulfilling all the conditions, or "`NIE`" otherwise. If the first line contains the word "`TAK`", then in the next m lines you should print the numbers $y_i \in \{0, 1\}$, such that $y_i \geq x_i$ and the graph formed from the edges with $y_i = 1$ should satisfy all the Witcher's conditions.

Example

standard input	standard output
2 4 5 1 2 0 2 3 0 3 1 0 1 4 1 2 4 1 5 4 1 2 0 2 3 0 3 4 1 4 5 0	TAK 1 0 0 1 1 NIE

Problem C. What a sequence!

Input file:	*standard input*
Output file:	*standard output*
Time limit:	2 seconds
Memory limit:	512 mebibytes

Let a_n be a sequence defined by the recursive formula:

$$a_{n+2} = k \cdot a_{n+1} + a_n$$
$$a_0 = 0$$
$$a_1 = 1$$

Given a certain $k \in \{1, 3, 5, 7\}$ and an odd prime number p, your task is to find the value of $a_p \bmod p$.

Input

In the first line one integer $Z \leq 10^6$ is given, denoting number of testcases described in following lines.

For each test case, first and the only input line contains two natural numbers p and k, p being an odd prime number.

$k \in \{1, 3, 5, 7\}$. The total length of the numbers p in the all testcases doesn't exceed 10^6.

Output

For each test case you should print exactly one line containing the value of $a_p \bmod p$.

Example

standard input	standard output
3 3 5 11 1 13 3	2 1 0

Problem D. Two Kilers

Input file:	*standard input*
Output file:	*standard output*
Time limit:	10 seconds
Memory limit:	512 mebibytes

In famous polish movie Two Kilers, main character Jurek Kiler receives regular deliveries of gold boxes. Each delivery can be described by value of gold boxes inside it, value of i-th delivery is equal to c_i.

Recently Jurek has felt concerned about falling value of deliveries. To check if those values are truly falling, he invented a measurement for sequence of deliveries, he calls it a *disorder* of a sequence.

For sequence of integers $c_1, c_2, \ldots, c_n$ the disorder of a sequence is equal to the length of the longest strictly increasing subsequence. Formally it is the largest integer s, such that there is an increasing sequence of indices $i_1 < i_2 < \ldots < i_s$ fulfilling $c_{i_1} < c_{i_2} < \ldots < c_{i_s}$.

Jurek thinks that his concerns are irrational if disorder of a sequence is at least k – then he is not interested in exact value of disorder. Problem is that it often turns out that some gold boxes from delivery are stolen or sometimes few gold boxes arrive with huge delay, so values of deliveries can change. You task is to calculate value of disorder after each change, or write k if disorder is at least k.

Input

In the first line one integer $Z \leq 10$ is given, denoting number of testcases described in following lines.

The first line contains two integers n, k, denoting number of deliveries and value of parameter k. The following line contains n integers $c_1, c_2, \ldots, c_n$, denoting initial values of the deliveries. The third line contains one integer q, denoting number of times delivery values change. Following q lines contains two integers p_j, v_j, denoting that $c_{p_j} = v_j$.

$n, q \in [1, 10^5]$, $c_i, v_j \in [1, 10^9]$, $p_j \in [1, n]$, $k \in [1, 20]$

Output

You should output q lines, each containing one natural number. If disorder value after j-th change equals s, then you should write $\min(s, k)$ in j-th line.

Example

standard input	standard output

Problem E. All in good fun!

Input file:	*standard input*
Output file:	*standard output*
Time limit:	1 second
Memory limit:	512 mebibytes

Policeman Biteusz loves pranking his friends from the police station. For example, his last joke was signing up his friend Bajteusz for patrolling the city on all Sundays and holidays left this year. However, Biteusz hasnt been so happy since he heard about the new work. Biteusz have to patrol three designated streets. Roads in Bajtocja are straight and no two streets are parallel, so he will have to patrol some, perhaps degenerate to a point, triangle. The policeman will stand at the best place possible, so to minimalize the maximum distance, which he'd possibly have to travel, when called to run to the specified street. Biteusz has been wondering how far he will have to run when called. Help him and calculate the minimum distance!

Input

In the first line one integer $Z \leq 10^4$ is given, denoting number of testcases described in following lines.

The first line of the each test case contains 3 — the number of roads in Bajtocja. Each of the next 3 lines contains three integers a_i, b_i, c_i, the description of the i-th road, meaning that the road is a line fulfilling the equation:

$$a_i x + b_i y + c_i = 0$$

$|a_i|, |b_i|, |c_i| \leq 10^6$, $|a_i| + |b_i| > 0$. No two given lines are parallel.

Output

For each test case you should print the minimum distance, which Biteusz will have to travel. Your answer will be accepted, if the absolute or relative error wouldn't exceed 10^{-6}.

Example

standard input	standard output
2 3 1 0 0 0 1 0 1 1 -1 3 1 1 -3 -3 1 1 1 0 -4	0.292893218813452 1.399173588432128

Problem F. Five Nights at Freddy's

Input file:	*standard input*
Output file:	*standard output*
Time limit:	2 seconds
Memory limit:	512 mebiytes

Maria has finally got her dream job! She became a professional night guard in a Sky Tower. One of her responsibilities is reporting about unwanted activity in the building. Fortunately for her, she can do her job without even leaving her cozy chair. All she has to do is keep track of cameras that are placed in the skyscraper. To do so she must handle n different cameras (enumerated from 1 to n). She wants to create the cycle of transmitting in which she will process information from the cameras. She will look at transition from some camera for one minute and then switch to the next camera on the cycle. Each camera should appear at least once on the cycle. There are some additional constraints. No two neighboring occurrences of the camera i on the cycle can be more than a_i places apart. Fortunately for Maria, for each $1 \le i < j \le n$ at least one of a_i and a_j is multiple of the other.

If it is possible, help Maria to create such cycle with length not exceeding 10^6.

Input

In the first line one integer $Z \le 50$ is given, denoting number of testcases described in following lines.

The first line of each test case contains one integer n, denoting the number cameras in the Sky Tower.

Following line contains a description of cameras. i-th number denotes the parameter a_i of the i-th camera.

$n \in [1, 10^5]$, $a_i \in [1, 10^5]$.

Output

For each test case:

If it is impossible to construct such cycle, the first (and only) line of the output should consist of single word "`NIE`".

If it is possible to construct such cycle, in the first line output single word "`TAK`". The following line should contain description of a transition cycle. First number m denotes the length of the cycle. Next m numbers are numbers of the cameras on the cycle in order they appear on the cycle.

If there are several possible answers, print any of them.

Example

standard input	standard output
2 3 2 4 8 5 3 3 6 6 42	TAK 4 1 2 1 3 NIE

Problem G. Choreography

Input file:	*standard input*
Output file:	*standard output*
Time limit:	10 seconds
Memory limit:	512 mebibytes

To celebrate the sixteenth anniversary of the unification of Vertical Byteland and Horizontal Byteland, a great parade will pass through the streets of the capital. The court choreographer Byteleon has decided to honor this event with a special choreography of the new national dance: Byterek.

Byterek is danced by n^2 dancers who are arranged in a square formation of size $n \times n$. It consists of a sequence of *vertical* and *horizontal* phases. During the dance dancers freely swap places with each other, on one condition, that for the time of vertical phases swapping is allowed exclusively within dancer's own column and during horizontal phases – within one's own row.

The choreography will also carry an additional meaning which should please king Byteasar. Every dancer will have an assigned costume of a specific color, so that the formation creates an image seen from the royal balcony. Byteleon wants the image to initially look like the flag of Vertical Byteland, and at the end of the dance – the flag of Horizontal Byteland. Unfortunately this task seems to be too hard for him, especially that the dance should consist of as few phases as possible, not to bore Byteasar. Please, help Byteleon and write a program that generates phases of Byterek.

Input

In the first line one integer $Z \leq 100$ is given, denoting number of testcases described in following lines.

The first line of each test case contains one integer n, denoting the length of columns and rows of the formation. The next n lines, each containing n integers, describe the initial arrangement of the dancers. Every number in $[1, n^2]$ occurs only once in this description and represents the desired position of the dancer in the final arrangement. This means that the final arrangement is a table in which every row and column is sorted, just like in the sample tests.

$n \in [1, 500]$, sum of n over all test cases does not exceed 1000.

Output

For each test case, in the first line your program should output an integer k – the minimum number of phases of Byterek. Then it should output k descriptions of arrangements in subsequent phases. The description of one arrangement consists of n rows each consisting of n integers, where each integer in $[1, n^2]$ occurs exactly once. If $k > 0$, the first arrangement must be possible to reach from the one given in the input after one phase of Byterek. The next one should be possible to reach from the previous one etc. The last arrangement should be equal to a sorted table. The order of vertical and horizontal phases is arbitrary.

Example

standard input	standard output
3 2 2 3 4 1 3 9 2 7 8 1 4 6 5 3 2 1 2 3 4	2 2 1 4 3 1 2 3 4 3 2 7 9 8 1 4 5 6 3 2 1 3 5 6 4 8 7 9 1 2 3 4 5 6 7 8 9 0

Problem H. Cheat

Input file:	*standard input*
Output file:	*standard output*
Time limit:	2 seconds
Memory limit:	512 mebibytes

Eryk with his partner called "Synek" are planning next spoof. Usually they swindle foreingers by offering false banknotes on low exchange rate, so they have to drive a lot around country to avoid recognition.

But to do this, they need to find a city where they can build a base. Cities and roads in their homeland can be viewed as a directed graph with n vertices numbered with integers from 1 to n and specific property – there is a directed edge from vertex i to vertex $i + 1$ for each valid i.

Since they want to get back to base after each "trip" so they finds cycles really attractive. They decided to build a base in a city which lies on all cycles in their country. Because there can be multiple such cities, they asked you to write down all of them. If there is no cycle in the graph, they can build a base in any city.

Formally a cycle is a path starting and ending in the same city and visiting at least one other city (possibly multiple times).

Input

In the first line one integer $Z \leq 50$ is given, denoting number of testcases described in following lines.

The first line of the test case contains two integers n and m, denoting the number of cities and roads. Each of the following m lines two integers a_i, b_i ($a_i \neq b_i$), denoting that there is a directed road from a_i to b_i. There can exist more than one road from a_i to b_i.

$n \in [1, 500\,000], m \in [0, 500\,000]$, sum of n and sum of m over all testcases does not exceed $1\,000\,000$.

Output

For each test case your program should write the number of cities where Eryk and "Synek" can build base, followed by indices of those cities in ascending order.

Example

standard input	standard output
sample.in	sample.out

Problem I. Interest

Input file:	*standard input*
Output file:	*standard output*
Time limit:	5 seconds
Memory limit:	512 mebibytes

Bitgard is a thriving metropolis. At this point, it has exactly n crossroads (numbered from 1, to n), connected by m one way roads. Each road has some non-negative length. The only point of road intersections are crossroads, though there may be many tunnels and viaducts. It is known that each crossroad is reachable from the crossroad number 1, but it doesn't have to be true, that each crossroad is reachable from any other crossroad.

Bitgard expands and get wealthy so quickly, that some people became interested in robbing it's facilities. Two of such individuals are Bolek and Lolek. To achieve their goal they moved to the crossroad number 1, and now they want to rob the shop nearby some other crossroad. If they decide to assault shop nearby the i-th crossroad, to minimize the risk of being associated with each other, they will go to this crossroad by two edge-separated paths. If such paths exist, Bolek and Lolek want to know the minimum sum of lengths of such paths.

For each crossroad other than the first one, help Bolek and Lolek to determine this value or state that this is impossible to find two such paths.

Input

In the first line one integer $Z \leq 100$ is given, denoting number of testcases described in following lines.

The first line of the test case contains two integers n and m, denoting the number crossroads and the number of one-way roads in Bitgard.

Each of the m following lines contains a description of roads. i-th line consists of three integer numbers a_i, b_i and c_i ($a_i \neq b_i \in [1, n], c \in [0, 10^9]$), denoting that there is road starting at crossroad a_i connecting it to crossroad number b_i, which has length c_i. You can assume that for each ordered pair of crossroads there is at most one road connecting them.

Sums over the values of parameters n and m over all test cases do not exceed 10^5 and 10^6 respectively.

Output

The first and only line of the output for each test case should consist of $n - 1$ (possibly zero) numbers denoting results for all the crossroads except from the first one. If it is possible to find two edge-disjoint paths from vertex 1 to vertex i, then i-th number should be equal to minimum length of sum of such paths. In other case you should output `-1` for this crossroad.

Examples

standard input	standard output
3 3 4 1 2 1 1 3 2 2 3 3 3 2 4 2 2 1 2 1 2 1 0 5 7 1 2 4 2 3 3 1 3 8 3 5 3 4 5 2 5 4 7 1 5 1	7 6 -1 -1 15 -1 11
2 4 9 1 2 18 2 3 1 3 4 11 4 3 2 4 1 30 3 1 24 3 2 22 1 3 18 2 4 1 4 5 1 2 2 1 3 14 1 4 4 2 1 20 2 3 30	58 37 48 -1 46 -1

Problem J. Planet of the singles

Input file:	*standard input*
Output file:	*standard output*
Time limit:	2 seconds
Memory limit:	512 mebibytes

Ania and Tomek have recently fallen in love. They are texting each other all the time. We can assume that each such message is a binary sequence. Tomek's father, Maksymilian, is not very keen on this whole situation. He decided to capture one of the lovers message and change it to his desired message, both this messages have the same length. He has to do it as fast as he can, because he doesn't want to gain unnecessary attention. Maksymilian can make three types of operations:

1. change 0 to 1 on some fixed position,
2. change 1 to 0 on some fixed position,
3. swap the bits on two fixed neighbouring positions.

Each of this operations take some time. The first one takes t_0 seconds, the second one t_1 seconds and the third one t_s seconds. Father can use any operation multiple times (possibly zero) and would like to change the given message into the desired one as fast as possible. Help him!

Input

In the first line one integer $Z \leq 20$ is given, denoting number of testcases described in following lines.

To compress the size of the input, all the binary sequences are converted into hexadecimal ones, so you may assume that the length of each such binary sequence is divisible by four.

The first line of the each test case contains four natural numbers n, t_0, t_1, t_s meaning the length of the hexadecimal sequences and the times described in the task description. The second line contains the message father got and the third line contains the message that father is going to make. This sequences consist of only characters from the set {`0, 1, ..., 9, A, B, ..., F`}.

$n \in [1, 10^6]$, $t_0, t_1, t_s \in [1, 10^{12}]$

Output

The first and only line of the output for each test case should contain one integer meaning the minimal number of seconds needed to change the first sequence into the second one.

Example

standard input	standard output
2 2 1 2 3 F0 D5 3 1 1 1 201 110	4 3

Note

In the first example the decoded binary sequences are `11110000` and `11010101`.

Problem K. Shadow

Input file:	*standard input*
Output file:	*standard output*
Time limit:	1 second
Memory limit:	512 mebibytes

As we all know, Earth is a unlimited plane in a 3D space and the sun is just a tiny ball shining evenly in all directions, for simplicity represented as a single point in a 3D space. One day the Death Star comes. The Death Star is a sphere with radius R. Because it is really big, some area on Earth was covered by the shadow. Your task is to calculate the area of the shadow, given the initial positions of Earth, the Death Star and the sun.

Input

In the first line one integer $Z \leq 10^4$ is given, denoting number of testcases described in following lines.

The first line of each test case contains three integers X_s, Y_s, Z_s meaning the initial sun coordinates. The second line contains four integers X_d, Y_d, Z_d and R meaning the position of the center of the Death Star and its radius. The third and last line contains four integers A, B, C and D meaning that Earth is a plane consisting of points (x, y, z) such that $Ax + By + Cz + D = 0$. You may assume that Earth, the Death Star and the sun are disjoint (they have no common points) and that the answer is greater than 0 and not infinity.

$X_s, Y_s, Z_s \in [-10^4, 10^4]$

$X_d, Y_d, Z_d \in [-10^4, 10^4]$, $R \in [1, 10^4]$

$A, B, C, D \in [-10^4, 10^4]$

Output

For each test case, the first and only line of the standard output should contain one real number, meaning the area of the shadow cast by the Death Star on Earth. Your answer will be accepted, if the relative or absolute error will not exceed 10^{-6}.

Examples

standard input	standard output
2 0 0 0 0 0 5 3 0 0 5 -50 0 0 0 6 8 0 6 6 8 0 -180	176.714586764 572.555261117
1 3 1 2 5 -1 4 2 2 -1 1 -22	77.432452329

Problem L. Boss of all bosses

Input file:	*standard input*
Output file:	*standard output*
Time limit:	2 seconds
Memory limit:	512 mebibytes

Krzysztof J. is a well-known and respected mafia boss from Szczecin. Next week, he is arranging a meeting with his associates. For simplicity let's denote them by numbers $1, 2, \ldots, n$. They will go for sushi or bowling and then talk business over dinner. Organizing such event is not an easy task though, because the invited mafiosos have a rather fiery temper and they often disagree on many topics, which sometimes results in fights. That is why he asked for your help.

You know that exactly $n - 1$ pairs of mafiosos are friends and for every such pair you have determined the expected time of *reaching a direct agreement* between them. You have also built a graph defined by the relation of friendship. As it turns out, that graph is connected, so for the pairs of mafiosos that are not friends, the expected time of *reaching an agreement* is the minimum sum of times of *reaching direct agreements* between adjacent mafiosos on the path connecting them in the graph.

During the dinner mafiosos will seat on one side of a long table. The seats have numbers $1, 2, \ldots, w + 1$, where w is the width of the table. Krzysztof doesn't want to provoke any fights during the dinner, so he decided that if there is a pair of mafiosos with seat numbers i and j and their expected time of *reaching an agreement* is d, then $d \leq |i - j|$ must hold. Because of that it can happen, that some seats remain empty. Your job is to determine the minimum width of the table, at which the guests can be seated.

Input

In the first line one integer $Z \leq 10^6$ is given, denoting number of testcases described in following lines.

The first line of each test case contains single integer n, denoting the number of mafiosos. Each of the following $n-1$ lines describes a pair of mafiosos that are friends. Pair i is denoted by three positive integers a_i, b_i and c_i – the labels of mafiosos and the expected time of *reaching a direct agreement* between them.

$n \geq 3$, sum of n over all test cases does not exceed 10^6.

Output

For each test case your program should write the minimum width of the table.

Example

standard input	standard output
2 3 1 2 1 2 3 1 5 4 1 3 1 5 2 2 5 1 3 5 2	2 9